What Happens When Purchase Intentions Are Blocked: Evidence from a Large Coffee Chain*

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Abstract

Temporary supply disruptions can affect not only when consumers buy but also what they choose after access returns. We study 18 store closures at a large Chinese coffee chain during the Covid-19 pandemic. A triple-differences design compares post-reopening changes for episodes with high versus low predicted purchase incidence. Purchase-day frequency rises 0.58 percentage points more per day for high-predicted-incidence episodes, although differential pretrends make this evidence descriptive. Novelty-seeking—the share of purchased products not previously chosen by the consumer—responds 4.15 percentage points less for high-predicted-incidence episodes. Under parallel triple trends and comparable general closure effects, this differential identifies the incremental response to interrupting a likely purchase. Heterogeneity by prior novelty-seeking is consistent with goal persistence: the forgone purchase remains salient and focuses returning consumers on familiar products.

Keywords: Customer Engagement, Customer Reactivation and Retention, Variety Seeking, Goal-Directed Consumption

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1 Introduction

Consumers sometimes face a temporary store closure when they are likely to purchase. The natural benchmark is that such a disruption shifts demand over time: consumers delay purchases and resume their usual behavior once access returns. We study whether this benchmark fails by distinguishing episodes with high and low predicted purchase incidence during the closure. When a closure interrupts a likely purchase, it may affect not only when consumers buy, but also how they choose once they return.

This distinction raises two central questions. Do interruptions to likely purchases reduce subsequent purchase frequency, or do consumers return to their prior activity? Does the post-reopening product-choice response differ between high- and low-predicted-incidence episodes? These questions determine whether temporary constraints merely reallocate demand across time or also reshape consumer choice.

We address these questions in the context of a large-scale digital coffee retail platform, where we observe detailed panel data on individual purchases of 770,00 consumers, store-level product assortments and store operations across a network of 260 locations for 18 months. A key feature of the setting is the occurrence of temporary store closures during Covid-19 pandemic that interrupt consumers' ability to transact at specific locations. These closures are driven by external operational constraints that affect store availability over time and are not targeted to individual consumers. As a result, they generate plausibly exogenous variation in purchasing opportunities at the store level, while leaving the broader demand environment largely unchanged. Because closures affect all potential customers of a given store simultaneously, they create sharp, localized disruptions that allow us to track how the same consumers adjust their behavior once constraints are lifted.

A central challenge is that the closure does not bind equally for all consumers. We use pre-disruption purchase behavior to construct an ex-ante prediction of purchase incidence during the closure window. We then estimate a triple-differences design that compares (i) treated versus control consumers, (ii) before versus after reopening, and (iii) high versus low predicted purchase incidence. The resulting high-minus-low contrast isolates the incremental response associated with interrupting a likely purchase only under the identifying assumptions stated below; it is not the total treatment effect for high-predicted-incidence episodes.

Our analysis yields two main results. First, purchase frequency rises by 0.67 percentage points per calendar day for high-predicted-incidence episodes and by 0.09 points for low-predicted-incidence episodes. The high-minus-low DDD is therefore 0.58 points (SE 0.25; $p = 0.021$). Because its pretrend test rejects equality, we interpret this result as descriptive evidence of post-reopening recovery rather than a clean causal rebound.

Second, novelty-seeking—the share of distinct products in the basket that the consumer has not purchased before—rises by 3.13 percentage points for low-predicted-incidence episodes. The corresponding high-group effect is a statistically insignificant 1.02-point decline. The DDD is -4.15 points (SE 1.45; $p = 0.004$): after netting out the low-group closure response, interrupting a likely purchase shifts choices toward products already familiar to the consumer.

As a robustness check on the outcome definition, the appendix replaces novelty-seeking with new-product-seeking, which measures the share of purchased products that have appeared recently in the transaction data. The alternative measure produces the same qualitative conclusion; Appendix C.2 reports the estimates and discusses their interpretation.

One interpretation is a goal-based mechanism. When a likely purchase is interrupted, the underlying goal may remain active. Upon reopening, consumers may prioritize goal completion and direct purchases toward familiar products. Heterogeneity by pre-disruption novelty-seeking is consistent with this interpretation, although it remains descriptive.

Our paper contributes to the marketing literature along three dimensions. First, we study how temporary constraints interact with consumer intent to shape demand, highlighting a form of state dependence driven by persistent purchase goals rather than solely by past choices or habits. Second, we contribute to the literature on supply-side shocks by showing that their effects depend on whether the constraint binds for the consumer, emphasizing the importance of heterogeneity in exposure to disruptions. The closest applied benchmark is Levine and Hristakeva (2025), who study a temporary grocery strike and use predicted trip displacement to distinguish baseline demand changes from the effect of an interrupted shopping trip. We use the same basic logic of separating exposure from binding interruption, but shift the object of interest from store choice and trip counts to product exploration within a changing digital assortment. Third, we speak to the literature on product discovery by showing how temporary interruptions can change exploration even when demand levels recover quickly. More broadly, our results suggest that understanding demand under temporary constraints requires going beyond average responses and explicitly accounting for how disruptions interact with consumer intent. This perspective provides a framework for analyzing environments in which purchasing opportunities are intermittently constrained, and highlights the role of consumer goals in shaping choice.

2 Behavioral Mechanism and Micro-Foundation

2.1 Counterfactual Thinking and Goal Persistence

A growing literature in marketing and consumer behavior studies how consumers respond when intended purchases cannot be completed due to temporary constraints such as stockouts, unavailable products, or interruptions in access. The standard benchmark is that these disruptions primarily shift purchases across time, with consumers delaying consumption and resuming their prior behavior once constraints are lifted. Under this view, temporary supply-side disruptions affect the timing of purchases but do not fundamentally alter the underlying decision process. However, prior research in consumer psychology suggests that interrupted actions can generate persistent cognitive and motivational effects that extend beyond simple postponement.

Our framework builds on the literature on counterfactual thinking and goal persistence. Counterfactual thinking refers to the mental simulation of how events could have unfolded differently under alternative circumstances (Epstude & Roese, 2008). Importantly, this literature argues that counterfactual thoughts are not merely retrospective evaluations, but functional cognitive processes that shape future behavior by maintaining attention on unrealized goals and missed actions. When individuals are prevented from completing an intended action, they naturally construct upward counterfactuals such as "I would have purchased if the store had remained open." These mental simulations increase the salience of the interrupted goal and motivate subsequent goal-completion behavior.

This perspective is particularly relevant in our setting. Temporary store closures do not simply eliminate purchasing opportunities at random. For consumers with an active purchase intention, the closure interrupts an ongoing consumption objective. As a result, the disruption creates a counterfactual consumption episode in which consumers mentally represent the purchase they would have made absent the closure. We argue that this counterfactual process generates a form of goal persistence that shapes post-reopening behavior.

We argue that counterfactual thinking represents the cognitive mechanism through which interrupted purchase intentions remain behaviorally active. When consumers are prevented from completing an intended purchase, they mentally simulate the forgone consumption episode. This counterfactual representation maintains the original consumption goal in an active state, leading consumers to return after reopening with more focused and less exploratory purchase behavior.

Our mechanism is closely related to research on interrupted goals and implemental mindsets in consumer behavior. Prior work shows that unfulfilled goals remain cognitively active and continue to influence subsequent decisions until they are completed (e.g. Gollwitzer, 1990; Kruglanski et al., 2002; Masicampo & Baumeister, 2011). Similarly, research on shopping momentum demonstrates that consumers who enter an implemental mindset become more focused on goal completion and less deliberative in their subsequent choices (Dhar et al., 2007). In our context, consumers whose purchases are blocked by store closures may remain in a suspended implemental state during the disruption period. Once stores reopen, these consumers return with a more focused purchase objective, prioritizing completion of the interrupted consumption goal rather than exploring alternative products.

This mechanism also relates to the literature on stockouts and constrained choice. Fitzsimons (2000) show that consumers' responses to unavailable products depend critically on the strength of their commitment to the intended purchase. When commitment is high, constraints generate stronger behavioral reactions and alter subsequent decision-making. Our setting extends this insight by considering temporary store closures that interrupt not only access to a specific product, but the entire execution of a planned consumption episode. Rather than weakening demand, these interruptions may preserve or even intensify purchase intentions through counterfactual processing.

A recent cognitive theory of reasoning and choice provides useful language for this channel. Bordalo et al. (2026) argue that decision makers first recognize the kind of problem they face using contextual cues and past experiences, and that this categorization determines which attributes receive attention. Applied to our setting, a closure can change the post-reopening problem from ordinary exploration in a coffee assortment to completion of an interrupted purchase routine. The mechanism therefore does not require consumers' tastes or information to change. Instead, the same consumer may place more attention on familiar products because the closure makes the familiar consumption path the natural category through which the next purchase is represented.

The implications of this mechanism are central to our empirical findings. If consumers whose purchase intention is blocked engage in counterfactual thinking and maintain an active purchase goal, then closures should not substantially reduce subsequent purchase frequency. Instead, consumers should return once constraints are lifted and complete the intended consumption activity. At the same time, because post-reopening behavior is driven by goal completion, consumers should become less likely to deviate from the originally intended purchase path. Consequently, consumers whose purchase intention is blocked should exhibit lower product exploration and reduced novelty-seeking after reopening.

2.2 A Behavioral Micro-Foundation

Consider a consumer who visits a coffee store and chooses between familiar products and new products. Familiar products are items the consumer has purchased before, while new products are items the consumer has not yet tried. In a standard choice framework, the probability of choosing a new product depends on intrinsic preferences, variety-seeking motives, and idiosyncratic taste shocks.

Our key departure is that a temporary closure can alter the consumer's attentional state. For consumers with an active purchase intention, the closure does not simply remove a transaction opportunity. It interrupts an intended consumption episode. This interruption induces counterfactual thinking: the consumer mentally represents the purchase she would have made had the store remained open. The counterfactual does not necessarily make one specific product more attractive. Instead, it reactivates the consumer's familiar consumption routine.

This distinction is important. In a coffee setting, the interrupted goal is often not "buy product j" in a narrow sense, but rather "complete my usual coffee purchase." The relevant object of salience is therefore the familiar consumption path: the set of products, routines, and choices that the consumer has previously experienced. Counterfactual thinking increases the cognitive accessibility of this familiar path. When the store reopens, the consumer returns with a more focused consumption objective, placing greater attentional weight on familiar products and less relative attention on unfamiliar alternatives.

The model can be summarized in three steps.

First, before the closure, the consumer has a familiar consumption routine formed through past purchases. This routine defines a set of familiar products that are cognitively accessible and easy to choose. New products may still be attractive, but they require attention, discovery, and willingness to deviate from the routine.

Second, the closure blocks the consumer's intended purchase. For consumers who would have purchased during the closure period, this generates a counterfactual representation: "I would have bought my usual coffee if the store had been open." This representation keeps the familiar routine active during the disruption.

Third, after reopening, the consumer makes a product choice under distorted attention. Familiar products receive an additional salience weight because they are linked to the interrupted consumption goal. New products do not receive this salience boost because they were not part of the counterfactual consumption script. As a result, the probability of choosing a familiar product rises, and the probability of novelty-seeking falls.

The framework does not impose a signed prediction on purchase frequency after reopening. Because the blocked intention may lead to delayed goal completion, substitution, or discouragement, the purchase-incidence effect is an empirical object. The formal predictions instead concern novelty-seeking. First, at the population level, blocked intention should reduce novelty-seeking because counterfactual thinking increases the salience of old products. Second, this reduction should be stronger for low-novelty consumers, widening the novelty gap between high- and low-novelty types.

This framework clarifies the behavioral distinction between counterfactual thinking and goal-focused behavior. Counterfactual thinking is the cognitive process generated by the interruption. It makes the unrealized consumption episode mentally accessible. Goal-focused behavior is the observable consequence: consumers return to complete the familiar consumption routine, reducing exploration.

It also distinguishes the counterfactual-goal mechanism from the missed-exposure mechanism. Under the counterfactual-goal mechanism, consumers are aware of alternatives but pay less attention to them because the familiar routine is salient. Under the missed-exposure mechanism, consumers fail to explore because new products are less likely to enter their consideration set. Both mechanisms predict lower novelty-seeking, but they operate through different primitives: salience and motivation in the first case, information and awareness in the second.

Mapping to notation. The discussion above implies a simple modeling restriction. When a consumer's intended coffee purchase is blocked, the relevant counterfactual is not an arbitrary product draw. The consumer thinks about something she has purchased in the past: the old product, routine, or familiar consumption path that would have been completed if the store had remained open. We therefore remove the notation that assigns probabilities to new-versus-old counterfactual content and assume that counterfactual thinking works through the persistent salience of old products. The model begins with one consumer type and then introduces heterogeneity in the strength of this old-product salience.

2.2.1 Setup

Consider a consumer facing closure event e and post-reopening period t. Let

$$C_e \in \{0,1\} \tag{2.1}$$

indicate whether the consumer's focal store is closed during event e. Closure exposure can affect all treated consumers, regardless of whether they had an active purchase intention during the closure. Let

$$I_e \in \{0,1\} \tag{2.2}$$

indicate whether the consumer had a purchase intention during the closure window, meaning that she would have purchased from the focal store absent the closure. The blocked-intention state is

$$B_e = C_e I_e. \tag{2.3}$$

Thus, C_e captures baseline closure exposure, while B_e captures the incremental case in which an intended purchase is prevented from being completed.

Let $P_{et} \in \{0,1\}$ indicate whether the consumer purchases after reopening. Purchase probability is

$$\Pr(P_{et} = 1) = \Lambda\left(\alpha_{et}^P + \beta_P C_e + \kappa B_e\right), \tag{2.4}$$

where $\Lambda(\cdot)$ is the logistic CDF. The parameter β_P captures the baseline closure-exposure effect on purchase incidence, while κ captures the incremental effect of blocked intention. The model does not sign either parameter. If blocked consumers return to complete the intended purchase, then κ may be positive. If the counterfactual memory vanishes or offsetting forces dominate, then κ may be zero. If consumers substitute away or become discouraged by the interruption, then κ may be negative. Purchase incidence is therefore an empirical object rather than a formal prediction of the model.

2.2.2 One consumer type

Let $A^O_{et} \in \{0,1\}$ indicate whether at least one old product enters the consumer's post-reopening consideration set. Old products are products the consumer purchased before the closure. The key carryover parameter is $\lambda \geq 0$. It measures whether the counterfactual representation of the forgone purchase remains accessible after reopening. If $\lambda = 0$, the counterfactual memory trace vanishes and blocked intention has no post-reopening salience effect. If $\lambda > 0$, the old product or old routine represented during closure becomes more accessible after reopening. This formulation follows memory-based consideration models in which accessibility affects consideration and choice without necessarily changing primitive product evaluations (Nedungadi, 1990).

Old-product consideration is given by

$$\Pr(A^O_{et} = 1) = \Lambda\left(\alpha^O_{et} + \beta_O C_e + \lambda B_e\right), \tag{2.5}$$

where β_O captures the baseline effect of closure exposure on old-product consideration and is unrestricted. The blocked-intention effect on the log-odds of old-product consideration is therefore

$$\Delta^O \equiv \frac{\partial}{\partial B_e} \log\left(\frac{\Pr(A^O_{et} = 1)}{1 - \Pr(A^O_{et} = 1)}\right) = \lambda. \tag{2.6}$$

This replaces the earlier formulation in which the effect depended on the probability that the counterfactual was about a new product. Here, blocked intention increases the log-odds of old-product consideration directly because the counterfactual is assumed to be about something purchased in the past.

To connect consideration to observed novelty-seeking, let $N_{et} \in \{0,1\}$ indicate whether the purchased product is new to the consumer, conditional on purchase. After consideration is formed, product utilities are stable:

$$u_{jt} = v_{jt} + \varepsilon_{jt}. \tag{2.7}$$

Closure exposure and blocked intention do not enter v_{jt}. They affect observed choice by changing which products are cognitively accessible. In particular, because old and new products compete for limited attention and choice, greater old-product salience reduces the relative probability that a new product is chosen. A compact logit representation is

$$\Pr(N_{et} = 1 \mid P_{et} = 1) = \frac{\sum_{j \in \mathcal{N}_t} \exp(v_{jt})}{\sum_{k \in \mathcal{O}_t} \exp(v_{kt} + \lambda B_e) + \sum_{j \in \mathcal{N}_t} \exp(v_{jt})}, \tag{2.8}$$

where $\mathcal{O}_t$ is the set of old products and $\mathcal{N}_t$ is the set of products new to the consumer. Equation (2.8) is not meant to impose that utilities change. Instead, the salience term summarizes the increased accessibility of old products in the post-reopening choice process. It implies

$$\frac{\partial}{\partial B_e} \Pr(N_{et} = 1 \mid P_{et} = 1) < 0 \qquad \text{if } \lambda > 0. \tag{2.9}$$

Prediction 1 (Population-level reduction in novelty-seeking)**.** If counterfactual salience persists after reopening, $\lambda > 0$, blocked intention raises the log-odds of old-product consideration and reduces the probability of choosing a new product conditional on purchase. Therefore, in the empirical triple-differences design, the population-level displacement effect on novelty-seeking is negative:

$$\delta_N^D < 0. \tag{2.10}$$

If $\lambda = 0$, the counterfactual memory trace does not carry over, and the model predicts no blocked-intention effect on novelty-seeking through this channel.

2.2.3 Two consumer types

Now allow consumers to differ in intrinsic novelty seeking. Let

$$\tau \in \{H, L\}, \tag{2.11}$$

where H denotes high novelty-seeking consumers and L denotes low novelty-seeking consumers. Let ν^τ denote the consumer's baseline tendency to consider and choose new products, with

$$\nu^H > \nu^L, \tag{2.12}$$

consistent with the view that novelty seeking is an individual-difference tendency that predicts exploratory consumption and product switching (Van Trijp et al., 1996; Baumgartner & Steenkamp, 1996).

The strength of old-product salience is also type-specific. Low-novelty consumers are more likely to think about old products and routines when a purchase is blocked, so

$$\Delta^{O,L} = \lambda^L > \Delta^{O,H} = \lambda^H \geq 0. \tag{2.13}$$

Equivalently, blocked intention raises the log-odds of old-product consideration more for low-novelty consumers than for high-novelty consumers. A type-specific version of old-product consideration is

$$\Pr(A_{et}^O = 1 \mid \tau) = \Lambda\left(\alpha_{et}^O + \beta_O C_e - \nu^\tau + \lambda^\tau B_e\right). \tag{2.14}$$

The term $-\nu^\tau$ captures that high-novelty consumers are less oriented toward old products at baseline, while $\lambda^\tau B_e$ captures the type-specific increase in old-product salience induced by blocked intention.

For novelty-seeking, the same limited-attention logic implies the reduced-form log-odds representation

$$\log\left(\frac{\Pr(A^N_{et}=1\mid\tau)}{1-\Pr(A^N_{et}=1\mid\tau)}\right)=\alpha^N_{et}+\beta_N C_e+\nu^\tau-\lambda^\tau B_e, \tag{2.15}$$

where A^N_{et} indicates that at least one new product enters the consumer's consideration set. The negative sign on $\lambda^\tau B_e$ reflects the crowd-out of new-product consideration by persistent old-product salience.

Without blocked intention, the high-low gap in the log-odds of new-product consideration is

$$G_0=\nu^H-\nu^L. \tag{2.16}$$

With blocked intention, the gap becomes

$$G_1=(\nu^H-\lambda^H)-(\nu^L-\lambda^L) \tag{2.17}$$

$$=(\nu^H-\nu^L)+(\lambda^L-\lambda^H). \tag{2.18}$$

Because $\lambda^L>\lambda^H$, blocked intention widens the novelty gap between high- and low-novelty consumers.

Prediction 2 (Blocked intention widens the novelty gap)**.** If counterfactual salience persists and low-novelty consumers think more about old products, $\lambda^L>\lambda^H$, then blocked intention reduces novelty-seeking more strongly for low-novelty consumers than for high-novelty consumers. The high-low novelty gap increases by

$$G_1-G_0=\lambda^L-\lambda^H>0. \tag{2.19}$$

In an empirical specification with a low-novelty interaction, the model predicts a negative incremental low-type effect,

$$\delta^D_{N\times Low}<0, \tag{2.20}$$

because the old-product salience generated by blocked intention is stronger for low-novelty consumers. Equivalently, in a specification where low-novelty consumers are the omitted group, the incremental effect for high-novelty consumers should be positive.

3 Setting and Data

We study consumer behavior using transaction-level data from a large Chinese quick-service coffee chain, where purchases are conducted primarily through a mobile application. The data span June 2020 to December 2021 and include approximately 10.6 million transactions from 779,000 customers.

The platform operates a dense network of physical stores with 260 locations and offers a dynamic product assortment that varies across stores and over time. Product availability is highly heterogeneous since a given product appears in 169 stores on average over its lifecycle, but is available in only 59 stores on a given day. This substantial variation implies that consumers' exposure to products depends critically on where and when they interact with the platform. The data track individual purchase histories at a high frequency, including the timing of transactions, store choice and product-level decisions. Consumers repeatedly interact with a relatively small set of nearby stores, generating persistent patterns in both purchasing frequency and product choice. This structure enables us to construct measures of baseline demand, store dependence and product exploration at the individual level.

A central feature of the setting is the presence of temporary store closures, which occur at the store-day level and prevent all consumers from transacting at the affected location. These closures arise from external operational constraints associated with Covid-19-related restrictions and generate sharp disruptions in purchasing opportunities. Because closures are not targeted to individual consumers and affect all potential customers of a store simultaneously, they provide plausibly exogenous variation in access to the platform.

Specifically, we focus on 18 unexpected store closures and construct a matched treatment-control design. Treatment customers are defined as those whose preferred store, identified based on pre-closure purchasing behavior, is the store that closes. Control customers are defined as those whose preferred store is one of a set of matched stores that remain open. The matching procedure ensures comparability across treated and control stores. For each closure, we select up to five control stores with similar store set-up times and exclude stores that are already used as controls for earlier closures. We then assign customers to treatment or control based on their preferred pre-closure store. To guarantee that store assignment is meaningful, we restrict attention to customers with at least five pre-closure purchase days and a preferred-store purchase share of at least 80%. The resulting panel contains 40,148 member-event pairs and 321,184 member-event-period observations for purchase frequency, including 7,390 treated and 32,758 control member-events. The classifier assigns 2,041 treated member-events, or 27.6% of treated episodes, to the high-predicted-incidence group. Each product-choice outcome uses 99,644 observations because it is defined only for periods in which a purchase occurs. This setting generates rich identifying variation along three dimensions. First, we observe the same consumers before and after a disruption, allowing for within-consumer comparisons. Second, closures vary across stores and over time, providing cross-sectional and temporal variation in exposure to constraints. Third, consumers differ in their reliance on specific stores, creating heterogeneity in the extent to which a closure constitutes a binding constraint.

Figure 1 presents the raw closure shock in purchase frequency. Relative period 0 is the closure window. Purchase frequency falls sharply for treated customers during this window, confirming that closures create a binding local supply constraint. After reopening, treated purchase frequency returns close to the control path, so the main empirical question is not whether demand permanently collapses, but whether the interruption changes what consumers buy when they return.

Table 1: Treatment-control balance before closure

Variable	Control mean	Treated mean	Difference	SE
Member-events	32,758	7,390		
Unique members	32,758	7,390		
Predicted purchase-incidence probability	0.307	0.313	0.006	0.004
High-predicted-incidence share	0.269	0.276	0.007	0.006
Pre-period purchase frequency	0.050	0.046	-0.003***	0.001
Pre-period novelty-seeking	0.540	0.558	0.018***	0.006
Closure duration in days	16.9	18.2	1.3***	0.1

Notes: The unit is a member-closure event. Pre-period purchase frequency is averaged over relative periods -4 to -1. Pre-period novelty-seeking is averaged over pre-period windows in which the member purchased at least one product. Difference is treatment minus control. Standard errors are heteroskedasticity-robust difference-in-means standard errors at the member-event level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

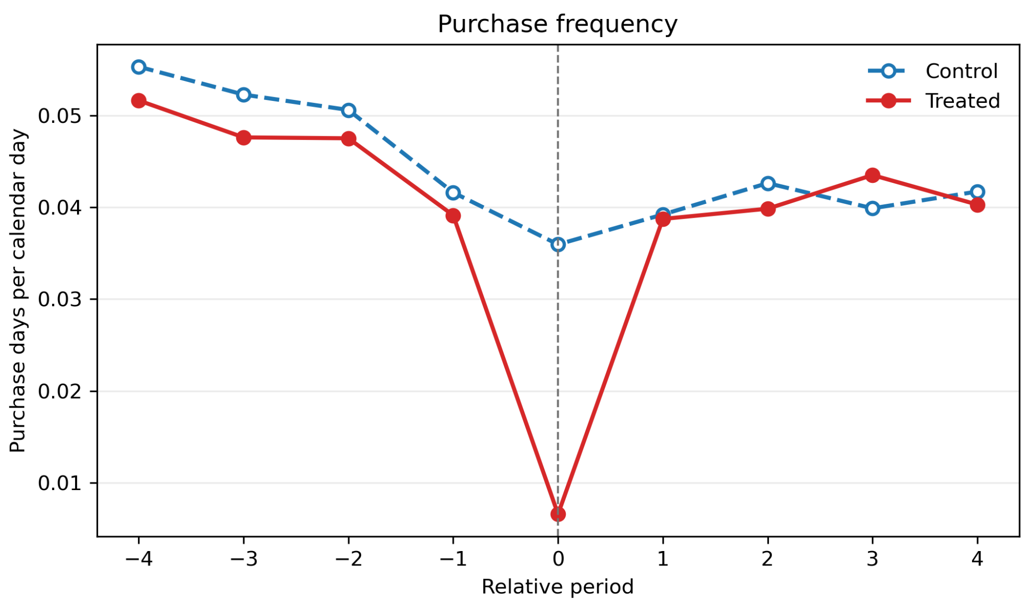

Figure 1: Raw closure shock in purchase frequency

Notes: The figure plots treatment and control mean purchase frequency by relative event period for the 18-closure analysis sample. Relative period 0 is the closure window. Purchase frequency is purchase days per calendar day.

Furthermore, the combination of repeated interactions, localized store dependence and time-varying product assortments creates an environment in which temporary disruptions can affect both the level and composition of demand. In particular, closures restrict purchasing opportunities and interrupt ongoing purchase processes. This structure allows us to study how temporary supply constraints affect consumer behavior along two margins: purchase frequency and product exploration. More broadly, the setting provides a natural laboratory to examine how consumers adjust when a planned purchase cannot be executed, and whether such interruptions have persistent effects on subsequent choices.

4 Empirical Strategy and Results

The central challenge in this setting is that not all treated customers are equally affected by a store closure. While all customers assigned to a treated store are exposed to the disruption, only a subset would have made a purchase during the closure window absent the interruption. Following the logic in Levine and Hristakeva (2025), we separate closure exposure from likely purchase interruption by classifying member-events according to predicted purchase incidence. A likely interrupted purchase occurs only when a high-predicted-incidence episode is also treated. The score is a prediction of purchase timing, not a direct measure of a psychological purchase intention.

4.1 Classifying Predicted Purchase Incidence

We classify each member-event pair with a predictive XGBoost model trained on pre-closure behavior. The model estimates whether the member would make at least one purchase in a window with the same length as the closure. We train one model for each closure duration, since the probability of at least one purchase mechanically depends on the number of days in the prediction window. Training uses relative periods -4 through -2, validation uses the holdout pre-closure period -1, and an additional audit evaluates control customers in period 0, where their matched stores remain open and actual purchase behavior is observed. For treated customers in period 0, the focal store is closed, so the outcome is counterfactual and we use the predicted probability. We classify a member-event as high predicted purchase incidence when this probability is at least 0.5.

The features use only information available before the prediction window: recent purchase frequency, recency since the last purchase, total pre-period purchase days, store loyalty, basket composition, discount and coupon use, day-of-week purchasing patterns, demographics and push-notification history.

Table 2 reports the validation results and the most important predictors. The classifier performs similarly for treated and control customers in the holdout pre-period, with accuracy around 77%. Accuracy is also similar for control customers in period 0, indicating that the model predicts purchase timing reasonably well in windows not used for training. The highest-ranked predictors are recent purchase intensity, total purchase history, recency and dependence on the preferred store. These patterns support the classifier's intended task: predicting purchase incidence during a closure-length window.

This distinction is central to our empirical design. The comparison between high- and low-predicted-incidence episodes separates general closure exposure from the additional response associated with a likely purchase interruption, subject to the identifying assumptions below. We focus on how temporary store closures affect consumer behavior along two dimensions: (i) purchase frequency and (ii) product choice, with a focus on consumers' propensity to explore new products.

4.2 Outcomes

Let i index consumers, e closure events, t time, and let L_e denote the duration in days of closure event e. For each event, we construct relative periods $t = -4, \ldots, 4$, where each period is a window of length L_e and period 0 is the closure window. The main regressions omit period 0 and compare four pre-closure windows with four post-reopening windows. This construction produces a member-event-period panel

Table 2: Purchase-intention classifier diagnostics

Panel A: Prediction performance

Evaluation sample	N	Accuracy	Precision	Recall	F1
Control, during closure period 0	36,766	0.752	0.524	0.544	0.525
Control, pre-period -1	36,766	0.766	0.638	0.559	0.592
Treated, pre-period -1	8,092	0.776	0.636	0.545	0.579
Overall weighted average	81,624	0.761	0.586	0.551	0.561

Panel B: Highest-ranked features

Predictor family	Mean importance	Median rank	Mean rank
Recent purchase frequency, last four weeks	28.49	1	1.23
Recent purchase frequency, last two weeks	11.54	2	7.00
Total pre-period purchase days	9.03	3	3.54
Days since last purchase	9.54	4	4.08
Preferred-store purchase share	3.38	7	7.31
Average pre-period purchase frequency	3.26	7	13.00

Notes: Panel A aggregates the duration-specific prediction audits using evaluation-sample weights. The period-0 control audit is possible because matched control stores remain open during the treated-store closure window. Panel B aggregates gain-based feature importance across duration-specific models and reports readable labels for the highest-ranked engineered features. Classifier audits are computed before final estimation restrictions.

in which each member-event contributes up to eight observations to the purchase-frequency analysis. We define two main outcomes on these windows.

Purchase frequency. We measure purchase intensity using purchase frequency, defined as the number of purchase days per calendar day within each event-length window. For consumer i and event e, we define purchase frequency as

$$\mathrm{Y}^{P}_{iet} = \frac{1}{L_e} \sum_{t \in \mathcal{W}_e} \mathbb{1}\{\text{Purchase}_{it} = 1\}, \tag{4.1}$$

where $\mathcal{W}_e$ denotes the set of days in the relevant window pre- or post-closure and $\mathbb{1}\{\cdot\}$ is an indicator equal to one if consumer i makes at least one purchase on day t. This measure captures the extensive margin of demand. Normalizing by closure length makes purchase frequency comparable across closure events of different durations.

Novelty-seeking. To measure exploration relative to a consumer's own purchase history, let $\mathcal{J}_{iet}$ denote the set of distinct products that consumer i purchases in event window t, and let $\mathcal{H}^{-}_{iet}$ denote the products the consumer purchased before that window. We define novelty-seeking as

$$\mathrm{Y}^{N}_{iet} = \frac{\left|\{j \in \mathcal{J}_{iet} : j \notin \mathcal{H}^{-}_{iet}\}\right|}{|\mathcal{J}_{iet}|}. \tag{4.2}$$

Each product counts once within a window, regardless of how many times the consumer purchases it. A higher value therefore indicates that a larger share of the consumer's distinct product set is new to that consumer. Novelty-seeking is defined only when a consumer purchases at least one product in the window; otherwise the denominator is zero and the outcome is missing. It therefore describes product composition conditional on purchasing, whereas purchase frequency captures the extensive margin. Appendix C.2 uses new-product-seeking as an alternative outcome definition based on products that have appeared recently in the transaction data.

Having defined the outcomes and predicted purchase-incidence status, we next plot the data for each group to visualize the variation behind the triple-differences design. Figure 2 plots average novelty-seeking separately across the four groups used for estimation, omitting the closure window. The left panel shows the patterns for low-predicted-incidence member-events. The open markers track control customers, and the solid markers track treated customers. Before the closure, the two groups have nearly identical novelty-seeking rates and no systematic gap. After reopening, treated customers are more likely than control customers to try products unfamiliar to them. Under the parallel-trends assumption, the difference-in-differences between these two series identifies the low-group closure effect.

The right panel compares treated and control customers with high predicted purchase incidence. Before the closure, treated high-predicted-incidence customers have higher novelty-seeking rates than control high-predicted-incidence customers, but the two series move similarly through most of the pre-period. After reopening, the treatment-control gap narrows as novelty-seeking falls more for treated high-predicted-incidence customers. This post-reopening difference-in-differences contains both the low-group closure effect and the additional high-minus-low contrast. The triple-differences specification below subtracts the left-panel difference-in-differences from the right-panel difference-in-differences.

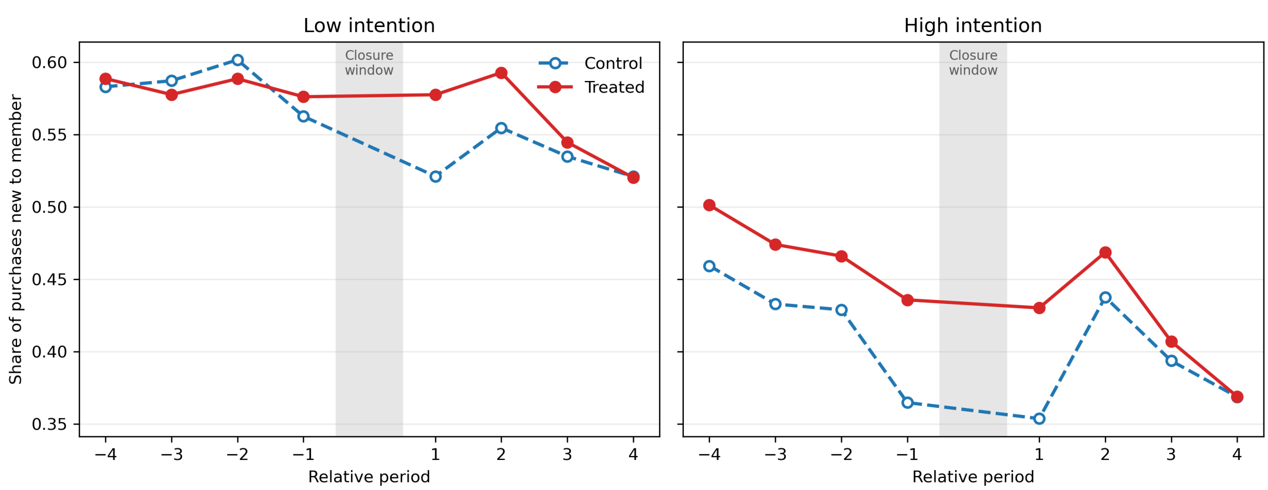

Figure 2: Novelty-seeking paths by predicted purchase intention

Notes: The figure plots treatment and control mean novelty-seeking by relative event period and predicted purchase-intention status for the 18-closure analysis sample. Novelty-seeking is the share of distinct products purchased in a window that are new to the consumer and is observed only in member-periods with purchases. Relative period 0, the closure window, is omitted because the main regressions compare pre-closure windows with post-reopening windows.

4.3 Triple Difference-in-Differences

To estimate the incremental response associated with interrupting a likely purchase, we compare (i) treated versus control customers, (ii) behavior before and after reopening, and (iii) high versus low predicted purchase incidence.

The logic follows the canonical triple-differences design in Gruber (1994), who studies group-specific mandates by comparing affected and unaffected groups across states with and without policy changes before and after adoption. In that setting, the third difference removes shocks common to adopting states, shocks common to the affected group, and aggregate time effects, leaving the component specific to the treated affected group. In our setting, closure exposure is the treated-unit dimension, predicted purchase incidence is the affected-group dimension, and the post-reopening period is the time dimension. Recent applications use the same idea in richer interaction settings. Gehring and Schaudt (2026), for example, interpret the interaction between drought exposure and insurance rollout as a triple-difference-style estimator with continuous treatment dimensions and emphasize that the main threat is an omitted factor that moves with the interaction term. This perspective motivates the coefficient decomposition and pretrend tests below.

Formally, we estimate the following pooled specification:

$$Y_{iet} = \delta^B \cdot \text{Post} \times \text{Treated} + \beta \cdot \text{Post} \times \text{High} + \delta^D \cdot \text{Post} \times \text{Treated} \times \text{High} + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}, \quad (4.3)$$

where Y_{iet} denotes the outcome of interest for member i in period t of event e. The indicator Post captures periods after reopening, Treated identifies customers exposed to a store closure, and High identifies episodes with high predicted purchase incidence during the closure window. This indicator is defined ex ante for both treated and control member-events; only treated high-predicted-incidence episodes face a likely interrupted purchase. The specification includes member-closure fixed effects ϕ_{ie}, relative-period fixed effects ω_t and calendar-month fixed effects γ_m. Standard errors are clustered at the consumer level.

The low-predicted-incidence treatment effect is δ^B. The high-predicted-incidence treatment effect is $\delta^B + \delta^D$. Thus δ^D is the high-minus-low treatment-effect differential, not the total effect for the high group. The parameter β is a high-minus-low post shift common to treated and control episodes; it is not a treatment effect.

4.3.1 Identification Assumptions and Diagnostic Tests

The causal interpretation of the high-minus-low differential requires two assumptions beyond the usual implementation conditions. First, parallel triple trends must hold: absent the closure, the treated-control gap for high-predicted-incidence episodes would have evolved like the corresponding gap for low-predicted-incidence episodes.

Second, we require *comparable general closure effects*. A closure may affect customers for reasons other than interrupting a likely purchase—for example, by changing convenience, promotions, substitution opportunities or perceptions of the store. The low-predicted-incidence comparison captures these general effects. The high-predicted-incidence comparison contains the same general effects plus the consequence of a likely interrupted purchase. If general closure effects would otherwise be comparable across the two groups, subtracting the low-group response from the high-group response isolates the incremental response associated with the interruption.

Third, predicted purchase incidence and the matched controls must be valid. The classifier uses only pre-closure behavior, and the matched control stores provide the counterfactual local-store path for treated customers. Under parallel triple trends and comparable general closure effects, δ^D identifies the incremental response associated with interrupting a likely purchase.

We assess these assumptions in the same order. First, Table 1 compares treatment and control member-events before closure. Treated and control episodes are similar in predicted purchase-incidence probability and high-predicted-incidence share, though treated episodes have slightly lower pre-period purchase frequency, slightly higher pre-period novelty-seeking and longer closure windows. These differences are not by themselves evidence against the design, but they motivate the fixed effects and pretrend checks below.

Second, Figure 1 verifies the first-stage shock. Purchase frequency falls sharply for treated customers during the closure window, while the control path remains available. This confirms that the closures create the intended local interruption while keeping the product-choice outcome conceptually separate from purchase incidence.

Third, we test pre-period trends using the event-study version of the triple-difference specification. For purchase frequency, the average treatment pretrend does not reject equality ($p = 0.895$), but the high-minus-low pretrend rejects ($p = 0.002$). We therefore treat the positive purchase-frequency estimate as supporting evidence that purchase-day frequency recovers after reopening, not as the cleanest causal result. For novelty-seeking, the corresponding high-minus-low pretrend test does not reject ($p = 0.797$). Appendix B reports these diagnostics, including the pretrend test for the alternative new-product-seeking measure.

4.4 Main Results

Table 3 follows the decomposition in Levine and Hristakeva (2025): Panel A reports the treated-control post contrast separately for low- and high-predicted-incidence episodes, and Panel B reports their difference. This layout keeps the total high-group effect distinct from the DDD. Appendix Tables C.1 and C.2 report every coefficient in the binary and continuous-score specifications.

Purchase frequency. The low-predicted-incidence treated-control post contrast is 0.09 percentage points per calendar day (SE 0.07; $p = 0.180$). The high-group contrast is 0.67 points (SE 0.24; $p = 0.006$), or 13.6% of the pre-period mean. Their difference is 0.58 points per day (SE 0.25; $p = 0.021$), about 11.8% of the pre-period mean. The estimates therefore show no post-reopening purchase-frequency shortfall for the high-predicted-incidence group. They instead describe a relative rebound. The high-minus-low purchase pretrend rejects equality, however, so the DDD cannot cleanly distinguish an interruption-induced rebound from continuation of differential pre-closure dynamics.

Novelty-seeking. For low-predicted-incidence episodes, the closure raises novelty-seeking by 3.13 percentage points (SE 1.20; $p = 0.009$). This low-group contrast captures the product-choice response to

Table 3: Decomposition of the main triple-differences results

Effect	Purchase frequency	Novelty-seeking
Panel A. Closure effects within predicted-incidence group		
Low predicted purchase incidence (δ^B)	0.0009	0.0313***
	(0.0007)	(0.0120)
High predicted purchase incidence ($\delta^B + \delta^D$)	0.0067***	-0.0102
	(0.0024)	(0.0083)
Panel B. Incremental interruption contrast		
High minus low (δ^D)	0.0058**	-0.0415***
	(0.0025)	(0.0145)
Pre-period outcome mean	0.0493	0.4978
Observations	321,184	99,644
R^2	0.562	0.416
Within R^2	0.013	0.001

Notes: Panel A reports treated-control post contrasts within low- and high-predicted-incidence episodes. Panel B reports their difference, the triple-differences coefficient. Purchase frequency is purchase days per calendar day. Novelty-seeking is the share of distinct products purchased in a window that the consumer had not purchased before that window and is defined only for windows with a purchase. The high-group effects are estimated in algebraically equivalent parameterizations and have their own cluster-robust standard errors. Standard errors, clustered at the member level, are in parentheses. All regressions include member-closure, relative-period and calendar-month fixed effects and omit the closure window. R^2 includes the absorbed fixed effects; within R^2 is calculated after absorbing them. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

general closure exposure. The high-group effect is a 1.02-point decline (SE 0.83; $p = 0.217$), which is not statistically distinguishable from zero. The high-minus-low DDD is -4.15 points (SE 1.45; $p = 0.004$), or 8.3% of the pre-period outcome mean. Under parallel triple trends and comparable general closure effects, the interruption of a likely purchase therefore offsets the low group's increase in exploration. The result establishes a negative relative response, not a precisely estimated absolute decline in novelty-seeking for the high group.

Appendix C.2 checks whether this conclusion depends on defining novelty relative to the consumer's own history. The robustness analysis instead measures new-product-seeking using products that have appeared recently in the transaction data. Its estimates show the same qualitative reduction in exploration after a likely purchase is interrupted.

4.5 Robustness Checks

We further assess robustness by replacing the binary high-predicted-incidence indicator with the centered ex-ante predicted purchase-incidence score. The treatment-by-score interaction is positive for purchase frequency ($+0.0076$, SE 0.0037) and negative for novelty-seeking (-0.0452, SE 0.0221). A 10-percentage-point increase in predicted purchase incidence therefore changes the treated-control post contrast by approximately $+0.08$ percentage points per day for purchase frequency and -0.45 points for novelty-seeking. These continuous-score estimates reproduce the signs of the binary DDDs without imposing the 0.5 classification threshold. Appendix Table C.2 reports the complete specification, including the baseline treated-control contrast and the common post-score gradient. Appendix Table C.3 reports the corresponding binary and continuous estimates for new-product-seeking.

5 Why Does This Happen? Goal Persistence and Consumer Type

5.1 Blocked Goals Create Focused Rebound

We propose a mechanism based on the persistence of active purchase goals. When a store closure blocks an intended purchase, the underlying goal need not disappear. Instead, the interruption may leave the goal cognitively active, increasing its salience and shaping subsequent behavior once purchasing opportunities are restored.

This idea comes from a large literature in psychology showing that unfulfilled goals remain mentally active, trigger counterfactual thinking about the missed action, and intensify the desire to complete the interrupted task (e.g., Brehm, 1966; Clee & Wicklund, 1980; Fitzsimons, 2000; Epstude & Roese, 2008; Masicampo & Baumeister, 2011; Kappes & Oettingen, 2016). In our context, a consumer whose likely purchase is interrupted may return after reopening with a more focused objective, prioritizing completion over exploration. This mechanism is consistent with the negative high-minus-low novelty differential. It does not imply, and the estimates do not establish, a statistically significant absolute novelty decline for high-predicted-incidence episodes.

5.1.1 Implementation and Findings

A direct implication of this mechanism is that some consumers should continue to pursue their purchase goal even during the closure period. In particular, if the goal remains active, consumers may substitute toward other stores of the same chain in order to complete the intended purchase. In the data, we observe that a non-trivial fraction of treated customers make purchases at alternative stores during the closure window. Specifically, 519 treated member-events involve at least one same-chain purchase at a different store while the focal store is closed, corresponding to 7.0% of treated member-events.

To assess whether such substitution drives the results, we re-estimate the triple difference-in-differences specification after excluding all treated member-events with any during-closure purchases at alternative stores within the chain. This restriction isolates consumers whose purchase opportunities are more likely to be interrupted. For purchase frequency, the high-minus-low DDD remains positive ($\delta^D = +0.0052^{**}$), while for novelty-seeking it remains negative ($\delta^D = -0.0323^{**}$). These contrasts are attenuated relative to the baseline but preserve the same sign and statistical significance. They are consistent with, but do not prove, a goal-persistence interpretation.

5.2 Heterogeneity by Pre-Period Novelty

The goal-persistence mechanism predicts that the incremental interruption contrast should differ across consumer types. Consumers who were already less exploratory before the closure have a more familiar consumption path to complete when a likely purchase is interrupted. For these consumers, counterfactual thinking about the forgone purchase may make familiar products especially salient after reopening. In contrast, consumers with high pre-period novelty-seeking have a more exploratory routine, so the high-minus-low DDD may be less negative. This is descriptive heterogeneity, not a separate causal subgroup design.

To test this prediction, we split member-closure episodes by pre-period novelty-seeking. For each episode, we compute the mean novelty-seeking rate over the four pre-closure windows in which the member makes a purchase. We then define $H_{ie} = 1$ if this pre-period novelty mean is above the sample median, and $H_{ie} = 0$ otherwise. The heterogeneity regression augments the main novelty-seeking specification as follows:

$$\begin{aligned} Y_{iet}^{V} = \delta_L^B &\cdot \text{Post} \times \text{Treated} + \beta_L \cdot \text{Post} \times \text{High} + \delta_L^D \cdot \text{Post} \times \text{Treated} \times \text{High} \\ &+ \theta^B \cdot \text{Post} \times \text{Treated} \times H_{ie} + \theta^\beta \cdot \text{Post} \times \text{High} \times H_{ie} + \theta^D \cdot \text{Post} \times \text{Treated} \times \text{High} \times H_{ie} \\ &+ \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}. \end{aligned} \tag{5.1}$$

The omitted subgroup is low pre-novelty episodes. Thus δ_L^D is the high-minus-low DDD for low pre-novelty consumers, and θ^D is the increment in that contrast for high pre-novelty consumers. As in the main specification, member-closure fixed effects absorb time-invariant differences across episodes, while relative-period and calendar-month fixed effects absorb common time shocks. Standard errors are clustered at the member level.

Table 4 reports the two coefficients needed to interpret this heterogeneity. For low pre-novelty consumers, the high-minus-low DDD is -26.3 percentage points. The contrast is 49.2 points higher for high pre-novelty consumers. The sign reversal suggests that the overall negative novelty differential is concentrated among consumers whose pre-disruption behavior was already less exploratory, but it should be read as descriptive mechanism-consistent evidence.

Table 4: Heterogeneity by pre-period novelty-seeking

Parameter or effect	Estimate	SE
Low pre-novelty effect (δ_L^D)	-0.263***	0.017
High pre-novelty increment (θ^D)	0.492***	0.024
Observations	95,986	
Member-closure fixed effects	Yes	
Relative-period and calendar-month fixed effects	Yes	

Notes: The dependent variable is novelty-seeking. Low and high pre-novelty groups are split at the sample median of episode-level pre-period novelty among members with observed pre-period novelty. The low pre-novelty row estimates δ_L^D in equation (5.1); the high pre-novelty increment estimates θ^D. Both are high-minus-low DDD contrasts, not total high-predicted-incidence effects. Standard errors are clustered at the member level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

This heterogeneity evidence should be interpreted as mechanism-consistent rather than as a separate randomized test. Pre-period novelty types differ in baseline behavior by construction, and may also differ in purchase frequency or other dimensions of engagement. The table therefore does not prove that pre-period novelty type itself causes the response. It shows that the main novelty-seeking decline appears exactly where the goal-persistence mechanism predicts it should appear: among consumers for whom familiar products were already the dominant pre-closure consumption path.

6 Conclusion

This paper studies how consumers respond when a temporary store closure is likely to interrupt a purchase. Using panel data from a large digital coffee chain, we estimate a triple-differences design that compares high- and low-predicted-incidence post-reopening responses. Purchase frequency is 0.58 percentage points per day higher for high- than for low-predicted-incidence episodes after reopening. Because the corresponding pretrend test rejects equality, this estimate is descriptive evidence of recovery rather than a clean causal rebound.

The product-choice result is more credible under the design's pretrend diagnostics. Novelty-seeking responds 4.15 percentage points less for high- than for low-predicted-incidence episodes. The direct high-group effect is a statistically insignificant 1.02-point decline. Under parallel triple trends and comparable general closure effects, this DDD identifies the incremental response associated with interrupting a likely purchase. An appendix robustness check based on new-product-seeking yields the same qualitative conclusion when exploration is defined relative to products that have appeared recently in the transaction data.

A goal-persistence mechanism is one interpretation: an interrupted likely purchase may make familiar products more salient upon return. The descriptive pre-period-novelty heterogeneity is consistent with this account, but neither the main DDD nor the heterogeneity regression isolates the psychological channel from other reasons that an interruption may reduce exploration.

Our findings contribute to the marketing literature in several ways. First, we provide empirical evidence on the role of goal persistence and state dependence in shaping consumer behavior under temporary constraints. While prior work has emphasized habit formation and inertia, our results highlight the importance of actively maintained purchase goals and their behavioral consequences when disrupted. Second, we show that supply-side shocks can have distinct effects on the level and composition of demand, and that these effects depend critically on consumer heterogeneity. By focusing on consumers for whom the closure likely binds, we uncover behavioral responses that are masked in aggregate analyses. Furthermore, our study contributes to a broader understanding of product discovery by showing how temporary interruptions can have lasting effects on consumer choice, even when demand levels recover quickly.

From a managerial perspective, purchase-day frequency can recover while product discovery weakens. The novelty-seeking estimate is a relative effect, and the alternative new-product-seeking definition strengthens the evidence that exploration narrows after a likely purchase is interrupted. Firms should therefore monitor product discovery alongside post-disruption sales recovery. More generally, demand recovery is incomplete when customers return but narrow the set of products they consider.

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Part

Online Appendix

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A Data Construction and Measurement

This appendix records the construction details that are needed to interpret the empirical design but are too mechanical for the main text. The unit of analysis is a member-closure event. A member can enter the analysis only through a closure event for which her pre-closure purchase history identifies a meaningful focal store.

A.1 Closure Registry and Member-Event Sample

The closure registry starts from a store-day panel. A candidate closure is a consecutive spell in which a store records no transactions. The raw detection step identifies 101 zero-demand spells. Because many university and college locations have long closures that reflect campus operating calendars rather than short disruptions, the analysis excludes those locations. The maintained paper design then focuses on the 18-closure registry used for the headline estimates.

Table A.1: Sample construction for the 18-closure design

Step or object	Count	Description
Transactions	10.6 million	Raw purchase records from the coffee-chain transaction data.
Customers	779,000	Customers observed in the transaction data.
Stores	260	Physical stores in the operating network.
Raw zero-demand closure spells	101	Consecutive store-day spells with no transactions.
University or college closure spells excluded	60	Long campus-related closures removed from the main disruption design.
Non-university spells before final registry filters	41	Raw non-university closure candidates before the treatment-control registry restrictions.
Main retained registry	18 closures	Headline registry used in the paper. Closure durations range from 10 to 29 days.
Member-closure episodes	40,148	Final episode-level sample underlying the purchase-frequency panel.
Treated episodes	7,390	Members whose pre-closure preferred store is the closing store.
Control episodes	32,758	Members whose pre-closure preferred store is a matched open control store.
Treated high-predicted-incidence episodes	2,041	Treated episodes with predicted closure-window purchase incidence above the classifier threshold, equal to 27.6% of treated episodes.

Notes: The full transaction counts describe the raw data environment. The final episode counts describe the main 18-closure estimation sample. Treatment and control assignment use pre-closure preferred stores and the same minimum purchase and store-loyalty restrictions for both groups.

For each closure, treated members are customers whose preferred pre-closure store is the store that closes. Control members are customers whose preferred pre-closure store is one of the matched control stores that remains open. Preferred stores are defined using pre-closure purchases, and the sample requires at least five pre-closure purchase days and a preferred-store purchase share of at least 80%. These restrictions make store assignment behaviorally meaningful: the closure is a relevant local disruption only if the customer had repeatedly relied on that store before the event.

A.2 Outcome Timing and Variable Definitions

Each closure event defines relative periods $t = -4, \ldots, 4$. Every relative period has the same length as the closure, denoted L_e, and period 0 is the closure window itself. The main post-reopening regressions omit period 0 and compare four pre-closure windows with four post-reopening windows. This convention produces 321,184 member-event-period observations for purchase frequency. Product-choice outcomes are observed only when a member purchases in the corresponding window; both novelty regressions use 99,644 observations after the final estimation restrictions.

Table A.2: Key variables in the empirical design

Variable	Construction	Interpretation
Treated	Preferred pre-closure store is the closed store.	Exposure to the local store closure. This is not the same as having a purchase blocked.
Control	Preferred pre-closure store is a matched store that remains open.	Counterfactual local-store path for treated customers.
High predicted purchase incidence	Ex-ante predicted probability of at least one purchase during the closure-length window is at least 0.5.	Predicted closure-window purchase incidence. It is defined before observing post-reopening outcomes.
Likely interrupted purchase	Treated $\times$ High predicted purchase incidence.	The closure is predicted to bind: the customer is exposed to the closure and likely would have purchased during the closure window.
Post	Relative periods 1 through 4.	Post-reopening windows. Period 0 is the closure window and is omitted from the main DDD regressions.
Purchase frequency	Purchase days in the relative-period window divided by L_e.	Extensive-margin purchase intensity, normalized for closure length.
Novelty-seeking	Share of distinct products purchased in the window that the consumer had not purchased before that window.	Product exploration relative to the consumer's own history, conditional on purchase.
New-product-seeking	Share of distinct products purchased whose first observed sale occurs in the current or immediately previous event window.	Adoption of recently appearing products, conditional on purchase. First observed sale need not equal the formal catalog launch date.

Notes: The main text uses "high predicted purchase incidence" for episodes with a score above the threshold. A likely interrupted purchase occurs only for treated high-predicted-incidence episodes, because controls retain access to their focal store.

A.3 Predicted Purchase-Incidence Classifier

The classifier separates closure exposure from likely purchase interruption. It predicts whether a member would purchase at least once in a window with the same length as the closure. This target depends mechanically on window length, so the implementation trains one XGBoost model for each observed closure duration. Training uses pre-closure relative periods -4 through -2, validation uses the holdout pre-period -1, and an additional audit uses control customers in period 0 because their matched focal stores remain open. The period-0 audit is a validation exercise; the DDD uses the ex-ante predicted score.

Table A.3: Classifier implementation details

Component	Implementation
Prediction target	Indicator for at least one purchase in a closure-length window.
Model family	XGBoost binary classifier, estimated separately by closure duration.
Training windows	Relative periods -4, -3 and -2.
Holdout validation	Relative period -1, separately audited for treated and control customers.
Additional audit	Control customers in period 0, where the matched control store remains open and actual purchase behavior is observed.
Scored sample	Treated and control member-closure episodes in the ex-ante period-0 panel.
Classification rule	High predicted purchase incidence equals one if the predicted period-0 purchase probability is at least 0.5.
Feature families	Recent purchase frequency, recency, total purchase history, preferred-store loyalty, basket composition, discount and coupon use, day-of-week purchasing patterns, demographics and push-notification history.

Notes: The prediction-performance table in the main text reports accuracy, precision, recall and F1 for the main validation slices. The appendix table emphasizes timing and information sets, because these are the key design concerns for interpreting the classifier as ex ante.

B Identification Diagnostics

This appendix reports the formal diagnostics behind the identification discussion in Section 4.3.1. The main design has three dimensions: treatment exposure, post-reopening timing and predicted purchase incidence. The key identifying restriction is therefore a triple-difference restriction: absent the closure, the treated-control gap for high-predicted-incidence episodes would have changed like the corresponding gap for low-predicted-incidence episodes.

B.1 Pretrend Tests

Table B.1 reports joint tests from the event-study analog of the main DDD specification. Each test uses the three pre-reopening event-time coefficients relative to period -1. The average treatment pretrend tests whether treated and control consumers move similarly before the closure. The baseline treatment pretrend tests the treated-control pretrend among low-predicted-incidence episodes. The high-minus-low DDD pretrend tests the pre-period coefficients on the treated-by-high-predicted-incidence interaction, which is the event-study analog of δ^D. The continuous-slope test replaces the binary indicator with the predicted purchase-incidence score.

Table B.1: Pretrend diagnostics for event-study specifications

Outcome	Test	Restrictions	p-value
Purchase frequency	Average treated-control pretrend	3	0.895
Purchase frequency	Baseline treatment pretrend for low-predicted-incidence episodes	3	0.055
Purchase frequency	High-minus-low DDD pretrend	3	0.002
Purchase frequency	Continuous incidence-slope pretrend	3	0.007
Novelty-seeking	Average treated-control pretrend	3	0.186
Novelty-seeking	Baseline treatment pretrend for low-predicted-incidence episodes	3	0.477
Novelty-seeking	High-minus-low DDD pretrend	3	0.797
Novelty-seeking	Continuous incidence-slope pretrend	3	0.778
New-product-seeking	Average treated-control pretrend	3	0.177
New-product-seeking	Baseline treatment pretrend for low-predicted-incidence episodes	3	0.990
New-product-seeking	High-minus-low DDD pretrend	3	0.515
New-product-seeking	Continuous incidence-slope pretrend	3	0.721

Notes: The table reports joint Wald tests of pre-reopening coefficients in event-study specifications. Purchase frequency uses 321,184 member-event-period observations. Novelty-seeking and new-product-seeking use 99,644 observations because product-choice outcomes are defined only in windows with purchases. Standard errors are clustered at the individual level.

The pretrend diagnostics support the product-choice interpretation more strongly than the purchase-frequency interpretation. For purchase frequency, the average treated-control pretrend is flat, but the high-minus-low DDD pretrend rejects equality. This is why the purchase-frequency result is interpreted cautiously as descriptive evidence of recovery rather than as the paper's cleanest causal estimate. For novelty-seeking, the high-minus-low DDD pretrend does not reject, and the same is true when the binary indicator is replaced by the continuous predicted purchase-incidence score. New-product-seeking shows the same pattern.

C Additional Results

C.1 Complete Main Specifications

The main text reports the group-specific effects and DDDs that carry the economic interpretation. This section reports the full coefficient vectors from the underlying regressions. Table C.1 presents equation (4.3) for the two main outcomes.

Table C.1: Complete binary predicted-incidence specifications

	Purchase frequency	Novelty-seeking
	(1)	(2)
Post × Treated (δ^B)	0.0009	0.0313***
	(0.0007)	(0.0120)
Post × High predicted incidence (β)	-0.0344***	0.0381***
	(0.0011)	(0.0064)
Post × Treated × High predicted incidence (δ^D)	0.0058**	-0.0415***
	(0.0025)	(0.0145)
Pre-period outcome mean	0.0493	0.4978
Observations	321,184	99,644
R^2	0.562	0.416
Within R^2	0.013	0.001
Member-closure fixed effects	Yes	Yes
Relative-period fixed effects	Yes	Yes
Calendar-month fixed effects	Yes	Yes

Notes: This table reports every coefficient from equation (4.3). Post × Treated is the low-predicted-incidence treated-control post contrast. Post × High captures the high-minus-low post shift common to treated and control episodes and is not a treatment effect. The triple interaction is the high-minus-low DDD. The closure window is omitted. Novelty-seeking is conditional on a purchase in the window. Standard errors, clustered at the member level, are in parentheses. Table 3 reports the directly estimated high-group effects and gives detailed outcome definitions. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The common post shift, β, is large for both outcomes. High-predicted-incidence episodes have lower post-period purchase frequency and higher post-period novelty-seeking than low-predicted-incidence episodes even after pooling treated and control observations. These coefficients reflect the relation between predicted purchase timing and post-period behavior; they are not closure effects. The DDD removes this common high-minus-low movement and isolates the additional treated high-group response under the identifying assumptions. The algebraically equivalent group-effect parameterization in Table 3 then combines δ^B and δ^D with covariance-aware standard errors.

C.1.1 Continuous predicted-incidence score

To avoid relying on the 0.5 classification threshold, we replace High_{ie} with the mean-centered predicted purchase-incidence score $\widetilde{s}_{ie}$:

$$Y_{iet} = \delta_0^B \cdot \text{Post} \times \text{Treated} + \beta^S \cdot \text{Post} \times \widetilde{s}_{ie} + \delta^S \cdot \text{Post} \times \text{Treated} \times \widetilde{s}_{ie} + \phi_{ie} + \omega_t + \gamma_m + \varepsilon_{iet}. \quad \text{(C.1)}$$

Because the score is centered, δ_0^B is the treated-control post contrast at the mean predicted incidence. The coefficient β^S is the common post-score gradient, while δ^S is the change in the treated-control post contrast associated with a one-unit increase in predicted purchase incidence.

Table C.2: Complete continuous predicted-incidence specifications

	Purchase frequency	Novelty-seeking
	(1)	(2)
Post × Treated	0.0023***	0.0190*
	(0.0008)	(0.0102)
Post × Centered predicted-incidence score	-0.0568***	0.0533***
	(0.0017)	(0.0098)
Post × Treated × Centered predicted-incidence score	0.0076**	-0.0452**
	(0.0037)	(0.0221)
Observations	321,184	99,644
R^2	0.565	0.416
Within R^2	0.0182	0.0004
Member-closure fixed effects	Yes	Yes
Relative-period fixed effects	Yes	Yes
Calendar-month fixed effects	Yes	Yes

Notes: The predicted-incidence score is centered at its estimation-sample mean. Post × Treated therefore gives the treated-control post contrast at the mean score. The score coefficient is the post-period score gradient common to treated and control episodes. The triple interaction shows how the treated-control post contrast changes with the predicted-incidence score. All remaining conventions and outcome definitions follow Table C.1. Standard errors, clustered at the member level, are in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The continuous estimates preserve the binary results' sign pattern. Higher predicted purchase incidence is associated with a larger treated-control recovery in purchase frequency and a smaller treated-control post response in novelty-seeking. For a 10-percentage-point increase in the predicted-incidence score, the fitted DDD changes by $+0.08$ percentage points per day for purchase frequency and -0.45 points for novelty-seeking. The purchase-frequency pretrend caveat remains, while the novelty-seeking score-pretrend test does not reject equality.

C.2 Alternative Outcome Definition: New-Product-Seeking

The main novelty-seeking outcome defines exploration relative to the consumer's own purchase history. As a robustness check, we instead ask whether consumers select products that have appeared recently in the transaction data. Let F_j be product j's first observed sale date and let $\mathcal{W}_{et}$ and $\mathcal{W}_{e,t-1}$ be the current and immediately previous event windows. We define new-product-seeking as

$$\mathrm{Y}_{iet}^{NP} = \frac{|\{j \in \mathcal{J}_{iet} : F_j \in \mathcal{W}_{e,t-1} \cup \mathcal{W}_{et}\}|}{|\mathcal{J}_{iet}|}. \tag{C.2}$$

This measure is the share of distinct products purchased in the current window whose first observed sale falls in the current or immediately previous event window. It proxies for recently introduced products, but first observed sale need not be the formal catalog launch date. Like novelty-seeking, it is defined only in windows with a purchase.

Table C.3: Robustness to using new-product-seeking

	Binary high-incidence	Continuous incidence score
	(1)	(2)
Panel A. Group effects and incremental interruption contrast		
Low predicted purchase incidence	0.0080	–
	(0.0092)	
High predicted purchase incidence	-0.0337***	–
	(0.0069)	
High minus low DDD	-0.0417***	–
	(0.0115)	
Panel B. Complete regression coefficients		
Post $\times$ Treated	0.0080	-0.0033
	(0.0092)	(0.0078)
Post $\times$ Predicted incidence	0.0300***	0.0468***
	(0.0050)	(0.0077)
Post $\times$ Treated $\times$ Predicted incidence	-0.0417***	-0.0486***
	(0.0115)	(0.0175)
Pre-period outcome mean	0.1456	0.1456
Observations	99,644	99,644
R^2	0.393	0.393
Within R^2	0.0008	0.0007
Member-closure fixed effects	Yes	Yes
Relative-period fixed effects	Yes	Yes
Calendar-month fixed effects	Yes	Yes

Notes: Column (1) uses the binary high-predicted-incidence indicator; column (2) uses the mean-centered predicted-incidence score. In Panel A, the low- and high-incidence rows are directly estimated treated-control post contrasts, and the high-minus-low row is their DDD. These discrete group effects are not defined for column (2). In Panel B, Post $\times$ Predicted incidence is the common post shift across treated and control episodes; it is not a treatment effect. In column (2), a one-unit change corresponds to moving across the full predicted-incidence scale. Standard errors, clustered at the member level, are in parentheses. All regressions omit the closure window. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The binary specification yields a 0.80-point low-incidence effect and a 3.37-point decline for high-predicted-incidence episodes. Their difference is -4.17 percentage points (SE 1.15; $p < 0.001$), or 28.6% of the pre-period mean. The continuous-score specification leads to the same conclusion: a 10-percentage-point increase in predicted purchase incidence lowers the treated-control post contrast by 0.49 percentage points. The high-minus-low pretrend tests do not reject for either specification ($p = 0.515$ for the binary measure and $p = 0.721$ for the continuous score). The robustness check therefore supports the main novelty-seeking result using a definition tied to recently appearing products rather than the consumer's own history.